

Pomegranate Disease Detection Using YOLOv12: A Deep Learning Approach Across Five Classes

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Abstract— Pomegranate (*Punica granatum*) suffers fairly substantial economic losses due to fruit diseases, which decrease fruit quantity and quality. Though professional manual diagnosis is the customary approach for detecting fruit disease, such methods are time-consuming, tedious, suffer from subjectivity, and are therefore difficult to use on a large scale. An automated deep learning model using the You Only Look Once (YOLOv12) object detection architecture was proposed for identifying multiple diseases in pomegranate fruits. The proposed method classifies fruits into one of five classes: Alternaria, Anthracnose, Bacterial Blight, Cercospora Fruit Spot, and Healthy. Five versions of YOLOv12 were trained on 5099 images with bounding box annotations and wide-ranging data augmentation, with the same hyperparameters. The model performance was measured using precision, recall, mAP@0.5, mAP@0.5:0.95 metric, inference latency, and model size. YOLOv12 has various sizes with suffixes n, s, m, l, and x, and YOLOv12s turned out to be the best model satisfying all dimensions with the highest precision, recall, mAP (IoU=0.5), and mAP (IoU=0.5:0.95) of 93%, 94%, 97% and 86%, respectively. YOLOv12s also has a lightweight architecture that allows it to be used on edge devices and mobile devices. This enables near real-time and highly accurate detection, allowing for early intervention and preventing crop loss. This system can be used in precision agriculture as a systematic and scalable automated monitoring system for detecting diseases in pomegranate.

Keywords— Pomegranate fruit, Disease detection, Deep learning, Precision agriculture.

I. INTRODUCTION

India is among the world's biggest producers of pomegranates, through decades and centuries of widespread agricultural growth and cultivated proliferation. Maharashtra, Karnataka, Gujarat, and Andhra Pradesh all produce a higher amount of pomegranates than they can consume, meaning there is a surplus for export. However, bacterial and fungal disease takes a severe toll on both the volume of production and the quality of fruit in India. The most serious diseases are Bacterial Blight (*Xanthomonas axonopodis*), Anthracnose (*Colletotrichum gloeosporioides*), Alternaria Fruit Spot (*Alternaria alternata*), and Cercospora Fruit Spot (*Cercospora punicea*). These diseases result in both external quality deterioration and internal development, which lowers the

fruit's marketability and produces lower economic value for farmers and exporters. A transfer learning method for pomegranate growth detection using deep learning convolutional neural networks was set up in [1]. Though they created a good maturity classification, they did not continue with disease detection, a much more complex method, because of in-class variability, illumination, and background noise on the farm. Image processing was utilized for pomegranate bacterial blight identification, but this traditional approach lacked speed and integration scalability at the field level. Identification and diagnosis of diseases are mainly done by experienced visual diagnosis in the field. Manual diagnosis by farming professionals is subjective, extremely time-consuming, and not feasible on a large scale for orchards with thousands of crops. Manual classification is feasible; however, it has been demonstrated that deep convolutional neural networks were able to identify plant disease with leaf images with exceptional success in accuracy on various crop types.

Recent developments in object detection and computer vision demonstrate that deep learning technology is increasingly effective in precision agriculture activities like fruit detection, ripeness quantification, harvesting prediction, pest detection, and others. Particularly in the YOLO variants, the best-performing systems are object detection networks as opposed to image classification networks because they assist in localization and classification concurrently, which can be utilized within a farm monitoring system that operates in real time. Authors in ref [2] developed a standardized dataset of 5,099 labelled images of five varieties (classes) of pomegranate fruits—Healthy(1450), Bacterial Blight(966), Anthracnose(1166), Cercospora (631) in Fruit Spot, and Alternaria (886), thus being a deep learning model training benchmark for disease detection. While some of the other deep learning models from object detection structures, such as YOLOv5, YOLOv8, and YOLOv11, have been employed in this dataset in other crops, few of them use the latest YOLOv12 framework in pomegranate disease detection. Hence, this research adopts a YOLOv12 detection model that seeks to precisely detect numerous pomegranate fruit diseases in real-time and also seeks to determine the top-performing model variant (n, s, m, l, x) for an optimal trade-off between accuracy and time for ease of deployment feasibility in edge

devices such as drones, smartphones, and embedded systems to enable effective disease management. This system is useful for smart agriculture since it may be an automated, large-scale, and affordable disease management tool for farmers. Ref [3] highlighted that integrating machine vision with agricultural practices can significantly enhance automation, improve fruit quality assessment, and support precision farming through timely detection and decision-making systems.

II. LITERATURE REVIEW

Transfer learning is common in agricultural image processing due to a lack of sufficient labelled datasets. The authors use pre-trained CNN architectures for classifying pomegranate stages of growth, where it is effectively noted that they can recognize the stages of maturity; however, this research lacks the other standard of secondary detection—localization—that would support segmentation for disease occurrence. The authors proposed EfficientNet [4] and transfer learning for leaf spot determination in grapes, providing high precision but limited to classification efforts and not annotating bounding boxes either. While these findings acknowledge the value of transfer learning, they also note the problems it presents with real-time detection and localization. Authors in ref [5] described that CNNs outperform traditional machine learning methods in plant disease detection classifications. In ref [6], authors trained deep neural networks to classify between 13 different plant leaf diseases, an important step in early automated agricultural diagnoses. Recently, YOLO architectures have been implemented as state-of-the-art real-time fruit monitoring and disease detection methods. For instance, [7] created a variant of YOLOv11 called YOLO-MSNet to detect pomegranates in growth stages in field conditions, whereas [8] used YOLO11-PGM for lightweight growth detection of pomegranates. The difference with this paper is that these two studies attempt to detect pomegranates in growth stages instead of diseases, and attempt to observe growth based on genetics instead of disease identification. However, both pomegranate studies were based on a standardized dataset that allowed for similar application of object detection models and disease classification; however, none have been benchmarked against the latest YOLO versions with a comparative approach.

Other successful YOLO models implementations exist for disease detection with other crops. In ref [9] created YOLO-LeafNet, a generalized framework for diseases in multiple species with a strong augmentation ability. In ref [10] created a lightweight instance segmentation model that recognizes both citrus pest and ripening states. In ref [11] optimized YOLO for passion fruit growing environments in orchards. Authors [12] developed ORD-YOLO for citrus recognition of ripening states under complicated illumination. Similarly, [13] optimized YOLOv3 for apple recognition of various growing states. This shows the generalizability of such dynamic systems across different growing situations. Authors in ref [14] created a lightweight YOLO-based architecture for apple leaf disease detection that achieves both high accuracy and reduced computational expense. For apple recognition, authors [15] created VBP-YOLO-prune, which improves algorithm success while dealing with inclement weather through feature fusion and structured pruning. In ref [16] showed enhancements that relied upon YOLOv9s-based adjustments for pear recognition small target recognition problems. Finally, in ref [17] created the EADD-YOLO model, which uses a YOLOv5-based approach to optimize

efficiency and achieve faster inference without sacrificing accuracy during apple leaf disease detection.

These works together prove that contemporary versions of YOLO still surpass the CNN-only classifiers and that in the fields of agricultural automation, where speed is necessary, this distinction becomes even clearer. An extensive overview of deep learning techniques for plant disease detection was provided by the authors in ref [18], noting that CNNs and object detectors play an increasingly important role in smart agriculture.

A. Limitations of Other Works and Research Gaps

Despite the major advances in the area of fruit detection, there exist two major gaps. First, much research on pomegranate multi-disease detection is lacking since most of the available studies address pomegranate counting and maturity prediction instead of detecting multiple diseases. Second, there are a few studies that have compared the new generation of YOLO models like YOLOv12. Rather, most of the work still uses older versions, such as YOLOv5 or YOLOv8, with less investigation of their scalability and deployability for edge deployment.

B. Novelty of the Proposed Work

This work introduces a comparative study of various variants (n, s, m, l, x) of YOLOv12 for pomegranate multi-disease detection. The performance of these models was compared on important metrics such as precision, recall, mean average precision (mAP), inference time, and model weight, emphasizing their usability in a real-time agricultural context. In addition, the research underscores the need for pragmatism, making the models efficient and deployable on low-power platforms, such as drones and smartphones, to enable on-site deployment and scalability.

III. PROPOSED METHODOLOGY

A. Dataset used

The standardized data employed within this research comprises 5,099 images of pomegranate fruits, each with five classes having their respective bounding boxes: Healthy, Bacterial Blight, Anthracnose, Cercospora Fruit Spot, and Alternaria Fruit Spot, as shown in Figure 1. These datasets were split into training, validation, and testing sets based on a standard split widely accepted in deep learning tasks [19]. To make the models more generalizable and robust, some of the mentioned data augmentation methods were used, such as random rotation and flipping, scaling and cropping, color jitter (adjustments in brightness, contrast, saturation), and Gaussian noise injection. These preprocessions were essential in making the models able to effectively deal with variations in lighting, orientation, and occlusion, as experienced in real-world agricultural settings[20].

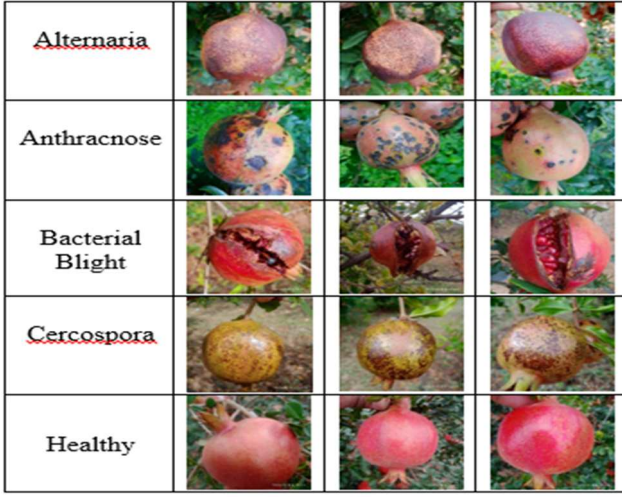


Fig.1. sample images from dataset

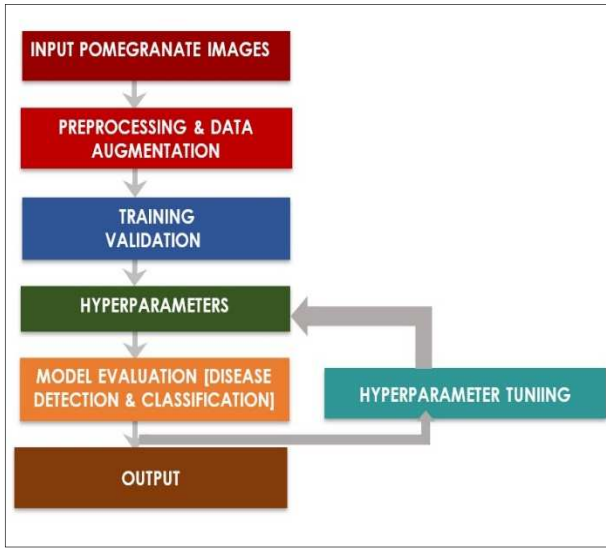


Fig.2. proposed work flow diagram

B. YOLOv12 Framework Overview

YOLOv12 represents the latest advancement in the YOLO object detection series, combining the speed of single-stage detection with enhanced accuracy through several key architectural features. These include a CSP-based backbone for improved feature extraction, a PANet-based feature pyramid for effective multi-scale object detection, CIoU loss for more precise bounding box regression, and Mosaic and MixUp augmentations to strengthen training performance. In this study, five YOLOv12 variants were evaluated—YOLOv12n, YOLOv12s, YOLOv12m, YOLOv12l, and YOLOv12x—to compare their effectiveness in pomegranate multi-disease detection. The YOLOv12n (Nano) model is the smallest, fastest, and least computationally demanding, offering slightly lower accuracy but achieving a strong mAP50 of 0.9698. The YOLOv12s (Small) variant provides the best balance between speed and accuracy, reaching an mAP50–95 of 0.8620, making it the most efficient all-round performer. The YOLOv12m (Medium) model is moderately larger and slower, offering improved robustness and balanced precision-recall over the nano variant. The YOLOv12l (Large)

version delivers high precision and recall with competitive accuracy, albeit with slower inference times. Finally, the YOLOv12x (Extra-Large) model, while possessing the greatest capacity, achieves a similar accuracy to YOLOv12l with a slightly lower mAP50–95 of 0.8180, and requires significantly more computational resources. All models were trained under identical hyperparameters to ensure a fair and consistent comparison across variants, as shown in Figure 3.

C. Model Training

The training procedure for YOLOv12 models was done by utilizing the Adam optimizer for learning at a rate of 0.001, which was regulated by a cosine annealing schedule for smooth convergence. The batch size was 16, and 100 epochs of training were performed, with the input image rescaled to 640×640 pixels in order to have a consistent scale across the dataset. The training utilized three most important loss functions: binary cross-entropy-based classification loss, objectness loss from IoU-based metrics, and bounding box regression loss from the Complete IoU (CIoU) formulation. Combined, these settings provided stable optimization, precise localization, and robust generalization performance for all model variants.

D. Evaluation and Performance Metrics

A rigorous object detection benchmark was performed using typical performance metrics to compare the YOLOv12 variants. The benchmark consisted of Precision (P), or the fraction of correctly detected objects out of all detections, and Recall (R), or the fraction of actual objects correctly detected by the model. In addition, two accuracy metrics were employed: mAP@0.5, the Mean Average Precision at an Intersection over Union (IoU) level of 0.5, and mAP@0.5:0.95, the mean of mAP at IoU levels from 0.5 to 0.95 in order to more fully cover the evaluation. Efficiency was measured by Inference Speed (FPS) to indicate real-time performance, while Model Size was evaluated to determine the viability of implementation on edge devices. Collectively, these metrics gave a comprehensive comparison of the five YOLOv12 versions in precision, efficiency, and scalability.

Precision (P):

$$P = \frac{TP}{TP + FP} \quad (1)$$

Recall (R):

$$R = \frac{TP}{TP + FN} \quad (2)$$

F1-Score:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3)$$

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

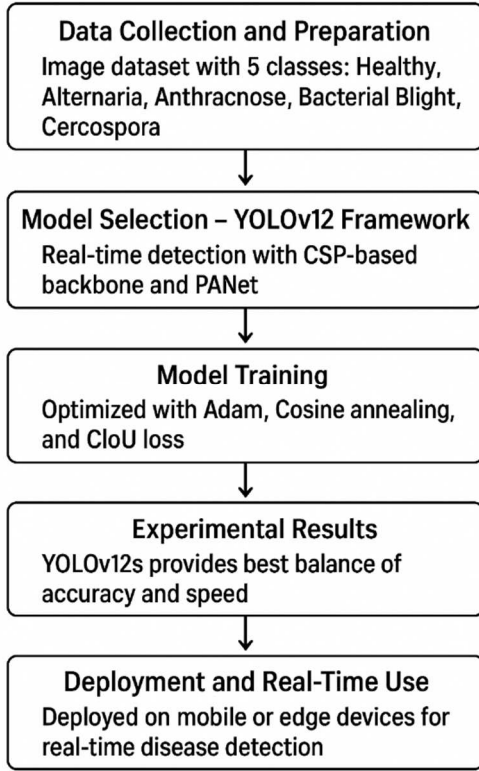


Fig. 3. Proposed model architecture

IV. EXPERIMENTAL RESULTS

This section presents a comprehensive evaluation of YOLOv12 object detection models on the dataset. We'll start with quantitative comparisons across all YOLOv12 variants, then dive into detailed visualizations of the YOLOv12 model. We chose this model as our representative because it strikes the best balance between precision and recall.

A. Quantitative Performance Comparison

We evaluated the entire YOLOv12 family using standard object detection metrics: Precision, Recall, mAP@0.5, and mAP@0.5:0.95. Table I shows results for all five YOLOv12 variants.

TABLE I. PERFORMANCE COMPARISON OF YOLOV12 VARIANTS:

Models	Precision	Recall	mAP50	mAP50-95
YOLOv12n	0.9153	0.9345	0.9698	0.8585
YOLOv12s	0.9300	0.9357	0.9695	0.8620
YOLOv12m	0.9238	0.9145	0.9656	0.8566
YOLOv12l	0.9280	0.9340	0.9690	0.8600
YOLOv12x	0.9040	0.8980	0.9480	0.8180

The evaluation results revealed several key observations regarding the performance of the YOLOv12 variants. In terms of precision and recall, the YOLOv12s model achieved the highest scores of 0.9300 and 0.9357, respectively, outperforming all other variants. For mAP@0.5, the

YOLOv12n model achieved a slightly higher value of 0.9698, marginally surpassing YOLOv12s, which recorded 0.9695. However, when considering mAP@0.5:0.95, the YOLOv12s model demonstrated superior performance with a score of 0.8620, indicating stronger accuracy across multiple IoU thresholds. Interestingly, model scaling did not always lead to better results — the largest model, YOLOv12x, did not consistently outperform mid-sized variants, suggesting potential overfitting or optimization inefficiencies in larger architectures.

B. YOLOv12s Model Visualizations

The YOLOv12s model demonstrates strong and stable training behavior across all visualizations. The training and validation loss curves show smooth and consistent convergence, with box, class, and DFL losses decreasing steadily as precision, recall, and mAP metrics rise across epochs, as shown in Figures 4 and 5. The smoothed metric plots further confirm this stability by reducing fluctuations and clearly depicting the model's improving detection accuracy over time.

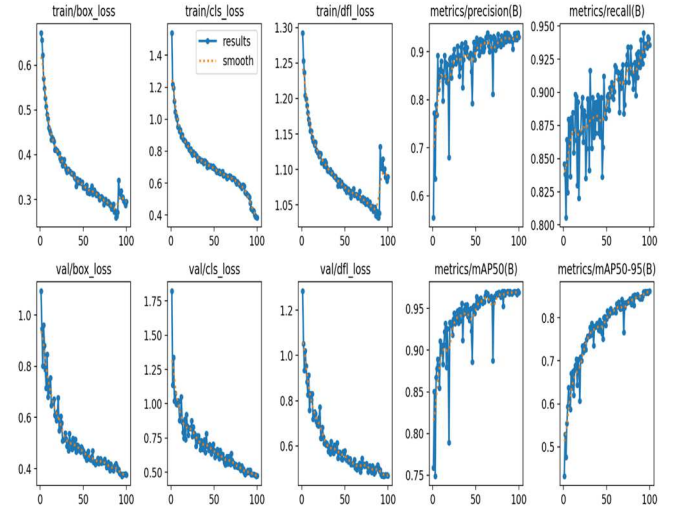


Fig. 4. Training performance graphs

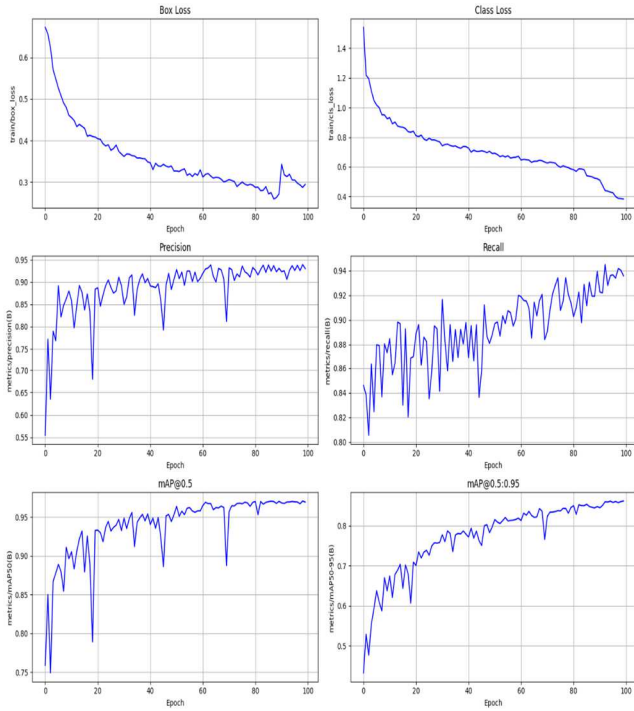


Fig. 5. Model performance over epochs and Training and validation loss

The confusion matrix reveals that the model accurately classifies most samples, with only minor misclassifications among visually similar disease categories, while the normalized confusion matrix highlights high per-class recall, particularly for Bacterial Blight and Cercospora, as shown in Figures 6 and 7. The per-class F1 scores, ranging from 0.88 to 0.98, reflect strong and balanced performance across all disease classes, confirming the robustness and reliability of the YOLOv12s model for plant disease detection as shown in Figures 8 and 9.

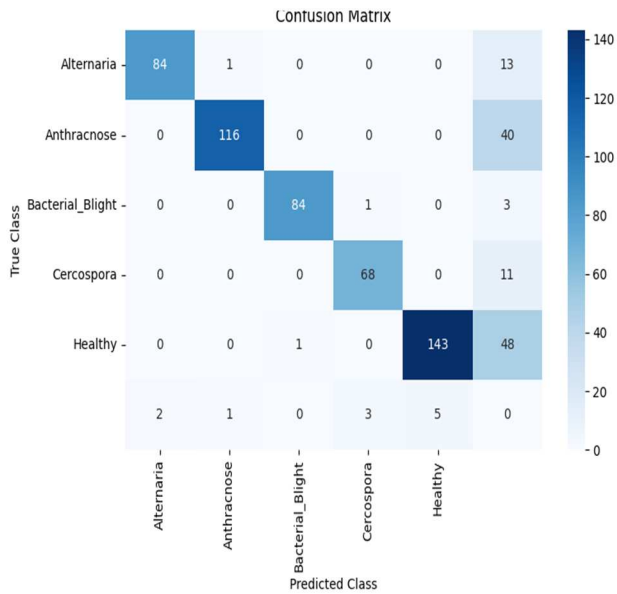


Fig. 6. Confusion matrix

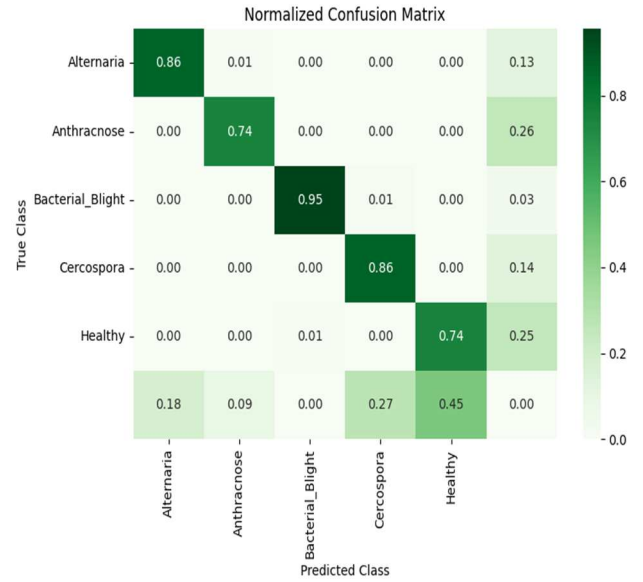


Fig. 7. Normalized Confusion matrix

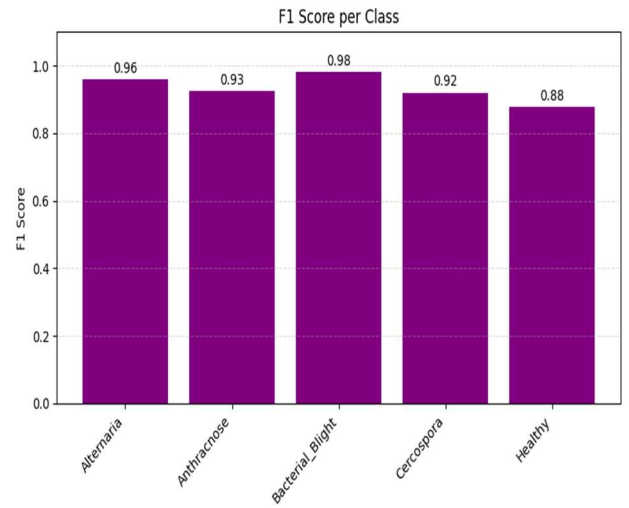


Fig. 8. F1 Score per Class

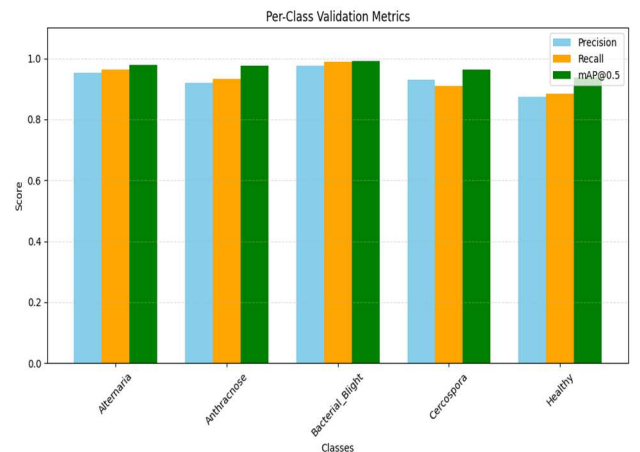


Fig. 9. Per class validation metrics

V. CONCLUSION

The experimental evaluation of YOLOv12 variants reveals how each model performs in object detection tasks. Testing five different models showed that YOLOv12s delivered the highest precision (0.9300) and recall (0.9357), plus the best

mAP@0.5:0.95 (0.8620). This combination suggests both accurate and thorough detection across the dataset. Mid-sized variants like YOLOv12l also showed strong performance, but the largest variant, YOLOv12x, fell short of expectations. This pattern suggests that bigger models don't automatically mean better results and can sometimes create inefficiencies or overfitting problems. Based on these findings, we chose YOLOv12s as our representative model for creating visualizations. These included training performance curves, confusion matrices, F1 scores, and per-class validation metrics. The training graphs display smooth convergence with losses dropping consistently, while precision, recall, and mAP values improved steadily across epochs. This progression confirms that detection quality kept getting better throughout training. The confusion matrices and per-class metrics show the model's solid performance, with high accuracy and consistent results across all object categories. The evaluation demonstrates that YOLOv12s strikes an effective balance between accuracy, efficiency, and model complexity. This makes it well-suited as a benchmark for real-time object detection applications. Combining quantitative results with visual analysis provides a comprehensive picture of how the model behaves, confirming its reliability and ability to generalize while offering direction for future work on lightweight, high-performing object detection systems.

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